EE 474 Introduction to Control Systems Homework 9

Due Wednesday, November 18, 2026

Reading:

- Franklin *et al.*: Chapter 5, Section 5.4 through end

Assignment :

1 System identification and design using Frequency Response information and Matlab invfreqs() command

- Necessary files:

  `HW9_InvfreqsExample.m`

Introduction

Step response can be used to identify a dominant pole or pole pair, but the data collected, such as t_r, M_p and t_s usually can not be used to identify plant zeros or additional plant poles.

By collecting frequency response data, the magnitude and phase of the response can be measured at each frequency tested, and several frequencies can be tested. From the magnitude and phase values at several frequencies, it is possible to estimate the locations of multiple poles and zeros.

Because a rational polynomial is a non-linear function, estimating the transfer function from frequency response data is a non-linear parameter estimation problem. Matlab's `invfreqs()` command solves this problem: it takes complex gain and frequencies in and returns the best fitting transfer function. To run `invfreqs`, the number of poles and zeros to fit must be specified.

File `HW9_InvfreqsExample.m` in the homework directory on the web site sets up an example illustrating the use of `invfreqs()`.

- Download file `HW9_InvfreqsExample.m` from the course web site
- Read `HW9_InvfreqsExample.m`, pay attention to the Matlab commands and the comments
- Read the Matlab help message

  >> help invfreqs

- Run `HW9_InvfreqsExample.m`. Simulated data are provided in the file. `HW9_InvfreqsExample.m` fits two transfer functions:
  - 2 poles, 0 zeros
  - 3 poles, 1 zero

Questions

A) Considering the frequency response plots, which transfer function better fits the data. (In statistical estimation, *residual* is a term applied to the errors left when the parameters are fit to the data. The residual appears in the plot title.)

B) Considering the pole-zero constellations, is each transfer function stable ?

C) Function `invfreqs()` has one substantial limitation: it does not determine the number of poles and zeros. Rather, the number of poles and zeros has to be put in as parameters `na` and `nb`. A typical procedure is to test several combinations of `na` and `nb`, and choose the one giving a physically feasible model and a good fit.

Modify the data in file `HW9_InvfreqsExample.m` to be:

```
Freq [Rad/sec]   Mag       Phase
       0.3142    0.4180   -6.0062
       0.6283    0.4237  -12.0334
       1.2566    0.4426  -24.2801
       3.1416    0.5135  -66.4484
       6.2832    0.2910 -131.1225
      12.5664    0.0754 -161.5733
      31.4159    0.0120 -173.4103
      62.8319    0.0030 -176.7621
```

This is the complex gain versus frequency for a system with transfer function

$$G_{uy}(s) = \frac{\widehat{B}(s)}{\widehat{A}(s)} \tag{1}$$

C.1) Determine $G_{uy}(s)$ for the models (indicate the transfer function):

```
Model1:  2 poles,  0 zeros
Model2:  3 poles,  1 zeros
```

C.2) Determine the better model based on these criteria:

- Is the model physically realistic:
  - We often have additional knowledge about the system. In this case we know that system $G_{uy}(s)$ is open-loop stable. Does the model have unstable poles ?

    If so, reject the model.
  - Does the model have poles or zeros that are substantially faster than the fastest data point, in this case

    $$\omega_{fastest} = 10\,\text{Hz} * 2\pi = 62.8\ [\text{rad/sec}]$$

    If so, reject the model (Poles and zeros outside the frequency range of the data will not be accurately determined; this helps determine which frequencies to test).

  - Is the model not physically feasible in some other way,
    If so, reject the model.

- Goodness of fit (does the model give a low residual)

2 Control Design:

For the system

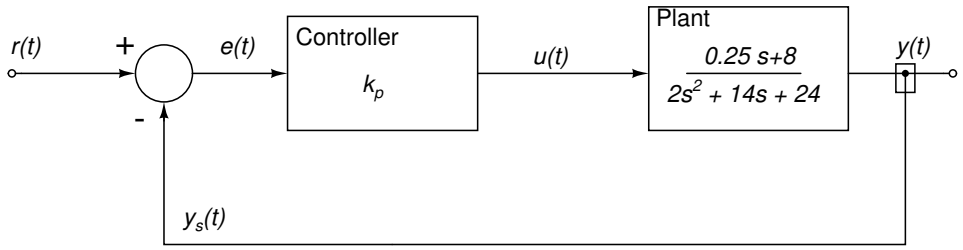


Figure 1: System with p-type control.

Test values of k_p on the range $k_p = 0.01 \ldots 100$. Select the value of k_p that best meets these design goals:

Rise time:	Less than 1/2 second
Settling time:	Less than 2 seconds
Steady-state error:	As small as possible
Overshoot:	Less than 10%

Note: Steady state error for a step input is defined as $e(t)$, after $y(t)$ reaches its steady value.

Tip: Setup the plant model as an LTI model in Matlab:

```
>> Guy = tf([.25 8], [2 14 24])
```

Choose a value of `kp`, such as

```
>> kp = 0.5
```

Build the closed-loop transfer function

```
%% The feedback() command applies the rule:
%% Forward path gain over 1 plus the loop gain.  The first argument
%% is the forward path gain, the second argument is any gain blocks
%% in the loop but not the forward path.
>> Try = feedback( kp * Guy, 1)
```

Plot the step response

```
>> step( Try )
```

Determine the step response parameters

```
>> stepinfo( Try )
```

Modify `kp` and test several values (This is the “Cut and Try” method of controller tuning).

Questions:

A) What is the suitable value of k_p ?

B) As k_p increases, does rise time increase of decrease ?

C) As k_p increases, does settling time increase of decrease ?

D) As k_p increases, does steady-state error increase of decrease ?

E) As k_p increases, does overshoot increase of decrease ?